

Supplementary Results — Module D

A structure-corrected two-locus screen for intrinsic incompatibilities

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This note reports the intrinsic (two-locus) arm. All results are structure-corrected but pre-simulation: the neutral, recombination-matched null (Methods, *Neutral, recombination-matched null*) is still required to convert the leads below into excess-over-neutral statements. Figures and tables are collected here.

1 The scan is calibrated and its strongest signal is not pair-specific

Treating each differentiated LD-reduced unit’s consensus dosage as a trait and testing it against every other unit in a double-LOCO, genome-PC-conditioned mixed model gives a calibrated scan (genomic control $\lambda \approx 0.98$ – 1.05 ; Methods, *Structure-corrected two-locus association*). Read out as one procedure — all 1.38×10^6 unlinked tests pooled at global false-discovery rate $q < 0.01$ (276 pairs), then paralogy-filtered (–29 pairs), region-merged (156 to 127 meta-nodes), and leverage-filtered (–50 edges) — the scan yields **110 associations among 83 meta-nodes** (18 trans, 92 cis; Table 1; Fig. 1).

The signal is dominated not by pair-specific incompatibility but by two removable or structural components. First, the loudest cross-chromosome associations are *technical*: pairs whose median within-population $|r|$ approaches one are assembly paralogues, corroborated by genotype concordance and excess heterozygosity (Methods, *Cross-chromosome paralogy*). Second, of the 110 surviving edges, **66 join two low-recombination “structure” meta-nodes** and only 13 join two candidate units (Table 1); the network is a small number of high-degree hubs (four with degree ≥ 10) sitting in the lowest recombination percentiles, rather than a field of isolated pairs (Fig. 1).

2 The structure module is individual-level admixture-block co-ancestry

The dominant object in the network is a single low-recombination complex whose members are wired almost entirely to one another. A principal-component analysis over the 87 member loci shows what it is *not* (Fig. 2). These loci separate individuals by population *less* than a random draw of differentiated loci: the mean per-locus among-population R^2 is 0.43, against 0.54 for random differentiated loci and 0.64 for the candidate loci. They also load only weakly on the genome-wide principal components ($|r| \leq 0.23$). Their cross-genome correlation is therefore neither among-population geographic differentiation nor the dominant ancestry axis, but individual-level admixture-block covariance: ancestral haplotype blocks co-inherited within individuals and, because these regions recombine slowly, not yet broken up. This is why the hubs persist through genome-PC conditioning, and it is read here directly from the genotypes rather than from the recombination map.

Table 1: Read-out of the structure-corrected scan. The pooled scan is reduced by one ordered procedure; the surviving edges are classified by the recombination annotation of their two endpoints (structure = recombination percentile < 0.10).

Stage	Pairs / edges
Unlinked (focal, partner) tests	1,381,804
Global Benjamini–Hochberg $q < 0.01$	276
– cross-chromosome paralogues ($ r _{\text{within}} > 0.9$)	247
Region-merged meta-nodes (from 156 units)	127
– single-population leverage ($ r _{\text{LOO}} < 0.3$)	110 edges
Final network	110 edges / 83 meta-nodes
trans (repulsion) / cis (coupling)	18 / 92
structure–structure / candidate–structure / candidate–candidate	66 / 31 / 13
hubs (degree ≥ 10) / structure-labelled nodes	4 / 59

3 The co-ancestry resolves into several cross-chromosome modules

The structure “module” is not one block but several. Average-linkage clustering of the meta-node consensus across chromosomes (cut at $|r| > 0.4$; Methods, *A single reproducible read-out*) resolves the network into 23 co-ancestry modules — a dominant module of 16 meta-nodes spanning 12 chromosomes and secondary modules of 14, 9 and 7 meta-nodes. Plotted one module at a time, each is a single coherent block: across all of a module’s SNPs on six or more chromosomes an individual takes one of the three dosages *together*, exactly as a co-inherited admixture haplotype block should, and unrelated to the other modules that the all-structure view mixes in (Fig. 3). Drawing only the within-module edges on the genome shows the same point: the co-ancestry resolves into a few disjoint, colour-coded bundles (79 of the 110 edges) rather than one hairball (Fig. 4).

4 The candidate residue

Thirteen edges join two candidate meta-nodes (Table 2; Fig. 5), and they organise into three kinds. (i) An aquilonia *co-sorting* module anchored on F33454 (Chr10; degree 12): its partners on Chr3/4/5/9/11/13/19 are almost all aquilonia-sorted, i.e. coordinated ancestry co-selection toward the minor (aquilonia) background rather than an obvious pairwise incompatibility — this module belongs with the ancestry-sorting arms. (ii) A single strong, ancestry-*independent* trans pair, F11431–F49480 (Chr3–Chr16; $q = 2.2 \times 10^{-10}$; both loci unsorted), the cleanest pairwise-incompatibility candidate the network contains. (iii) A few same-ancestry cis pairs (polychaeta: F17114–F49429, F15820–F26284; aquilonia: F22294–F39587), i.e. co-sorting, not repulsion. Stripped of structure and of ancestry co-sorting, the pair-specific residue is F11431–F49480 together with a couple of weak, leverage-stable trans pairs — a handful of edges reached by one reproducible criterion.

5 Low recombination is a caveat, not a verdict

The interpretation above is descriptive and stops short of an intrinsic call, for one reason. Low recombination is not evidence of neutrality: it is where coadapted complexes are expected to

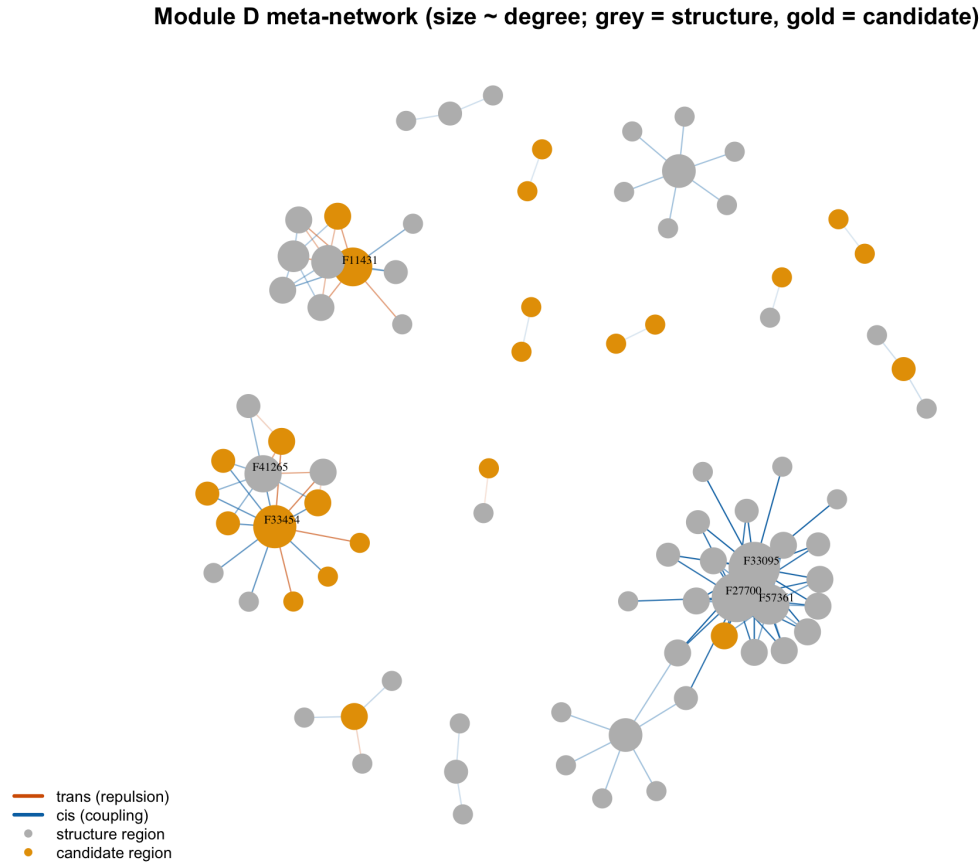


Figure 1: **The structure-corrected two-locus network.** Meta-nodes sized by degree; grey = low-recombination “structure” annotation, gold = candidate; blue edges cis (coupling), orange trans (repulsion). The network resolves into a large grey structure complex (the low-recombination hubs), a gold candidate hub (the F33454 co-sorting module), and scattered candidate pairs.

accumulate, so the “structure” annotation is used as context, never as a filter, and every unit — low-recombination included — is carried into the neutral null. Because for an unlinked pair the between-region recombination fraction is 0.5 regardless of local rate, low local recombination does not hold cross-chromosome LD; it makes each region a clean, low-noise ancestry marker that lights up for whatever residual covariation remains after conditioning, neutral or selected. The PCA above is consistent with neutral relic co-ancestry but does not exclude selection, which parallel or stabilising forms can also render weakly differentiated (Methods, *Recombination annotation, not a filter*).

The clean discriminant is a neutral null matched to *local* recombination (Methods, *Neutral, recombination-matched null*): every pair is judged against a baseline simulated at its own recombination rate, so a low-recombination pair can still surface if its LD exceeds that high baseline. Two corroborations are available before the simulation and both separate selection from neutral relic co-ancestry: replicate-population parallelism (neutral drift takes random directions across the twenty populations, whereas selection maintains the same combination reproducibly) and specific exceedance of a module over the general low-recombination background. The coherent multi-chromosome blocks (Fig. 3) are therefore the first coadaptation candidates to test, not the ones to set aside. Until the null gives them a scale, the intrinsic screen reports a largely negative result — the apparent two-locus signal is structure, assembly, or single-deme leverage — with a short, explicit list of candidates to resolve.

Structure-module PCA (87 low-recomb hub loci) -- individuals coloured by popu

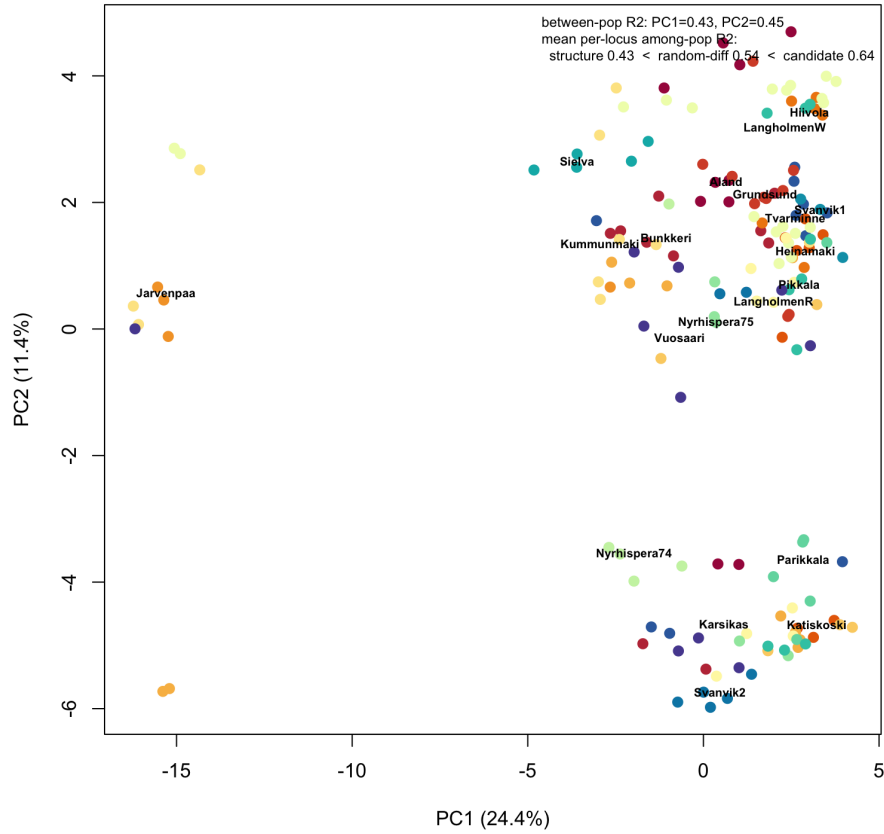


Figure 2: **The structure module carries less among-population structure than baseline.** PCA over the 87 loci of the structure module, individuals coloured by population. Real between-population structure is present (some populations, e.g. Järvenpää, form coherent clusters), but mean per-locus among-population R^2 (0.43) is below that of a random draw of differentiated loci (0.54) and of the candidate loci (0.64), and the module's axes correlate only weakly with the genome-wide PCs. The pattern is consistent with individual-level admixture-block co-ancestry rather than geographic differentiation.

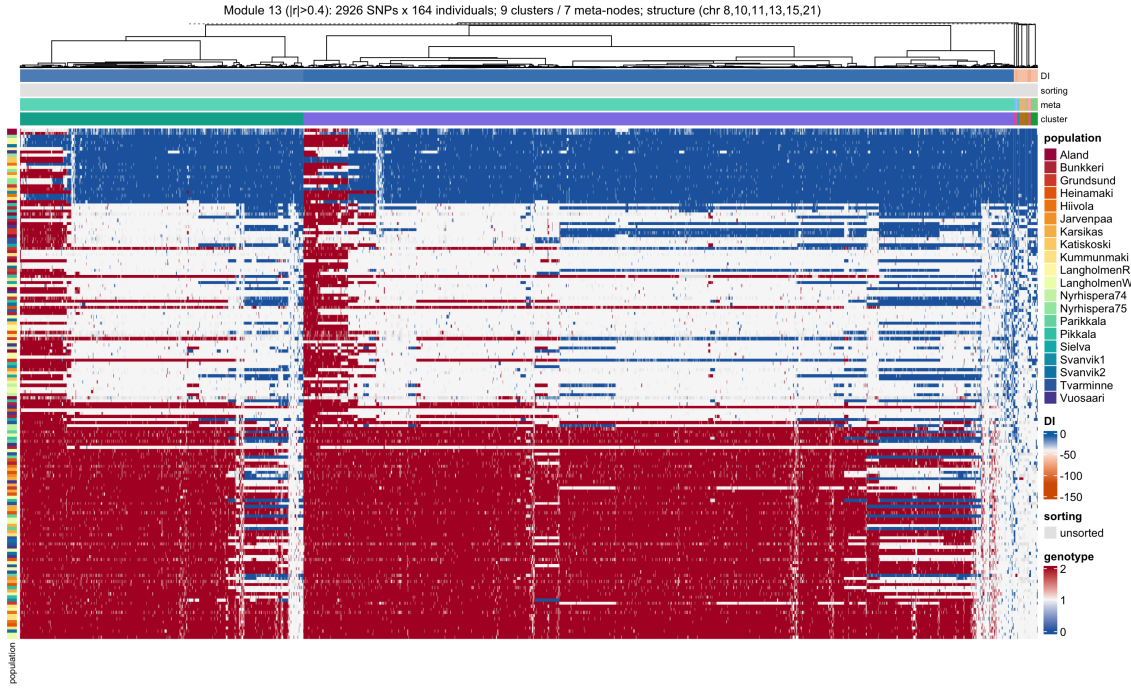


Figure 3: **One cross-chromosome co-ancestry module behaves as a single haplotype block.** Genotype heatmap of a module’s member SNPs (9 clusters / 7 meta-nodes across Chr8/10/11/13/15/21; columns clustered; colour bars: cluster, meta-node, sorting, DI; rows = individuals along the module’s co-ancestry PC1). An individual is homozygous-reference (blue), heterozygous (pale), or homozygous-alternate (red) consistently across all 2,926 SNPs on all six chromosomes, and the three groups do not coincide with populations.

Table 2: The 13 candidate–candidate edges (both endpoints outside the structure module), with chromosomes, coupling, false-discovery q , and the ancestry-sorting class of each endpoint. The F33454 anchor and its aquilonia-sorted partners form a co-sorting module; F11431–F49480 is the one strong ancestry-independent trans pair.

Pair	Chr	coupling	q	sorting (a/b)
F11431–F49480	3/16	trans	2.2×10^{-10}	unsorted/unsorted
F10974–F33454	3/10	trans	2.7×10^{-4}	aqu/aqu
F32802–F33454	9/10	trans	1.1×10^{-3}	aqu/aqu
F11265–F33454	3/10	trans	6.2×10^{-3}	pol/aqu
F15419–F33454	4/10	cis	2.8×10^{-11}	aqu/aqu
F33454–F57178	10/19	cis	8.8×10^{-5}	aqu/aqu
F33454–F41301	10/13	cis	1.2×10^{-4}	aqu/aqu
F33454–F35575	10/11	cis	2.2×10^{-4}	aqu/aqu
F17114–F49429	4/15	cis	9.2×10^{-4}	pol/pol
F15820–F26284	4/7	cis	2.3×10^{-3}	pol/pol
F21099–F33454	5/10	cis	3.0×10^{-3}	aqu/aqu
F22294–F39587	6/12	cis	4.7×10^{-3}	aqu/aqu
F45183–F45504	14/14	cis	5.0×10^{-3}	unsorted/unsorted

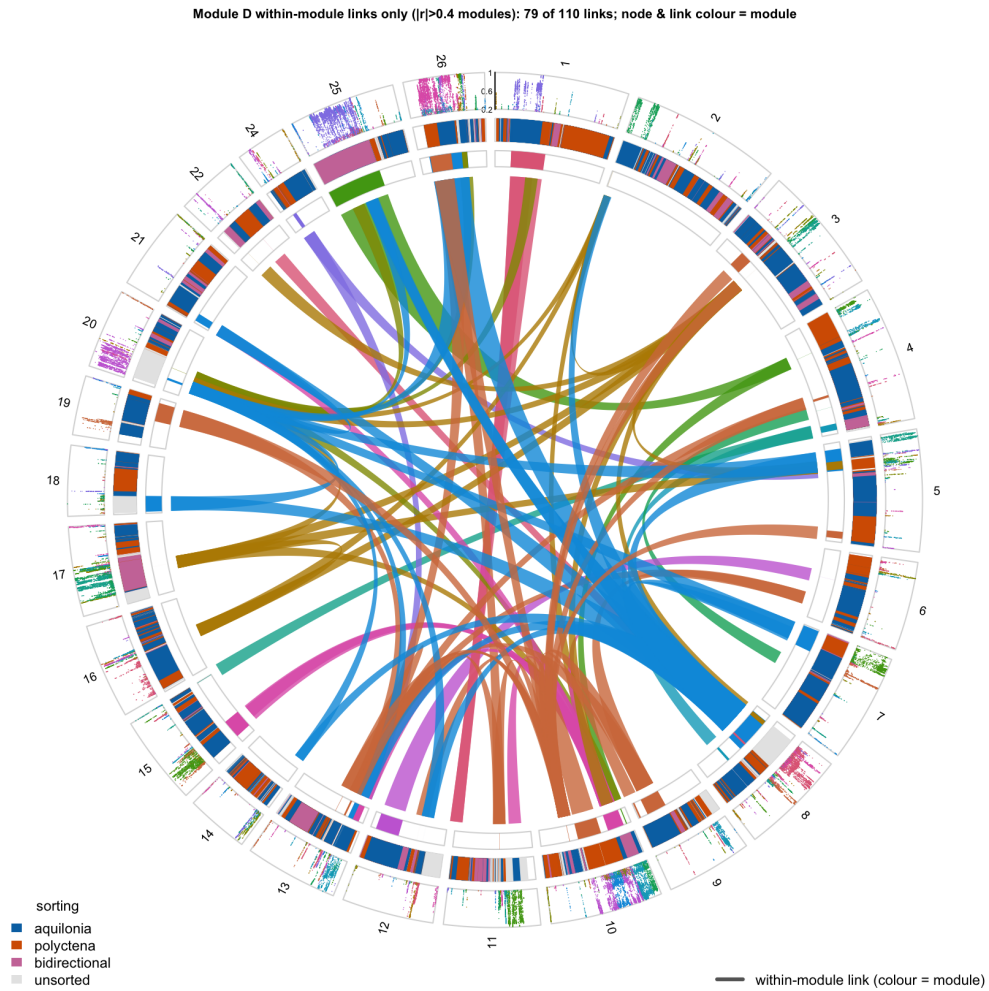


Figure 4: **Within-module associations.** Only edges whose two meta-nodes fall in the same cross-chromosome correlation module (79 of 110), coloured by module; node ring coloured to match. Each co-ancestry module reads as its own bundle of links across chromosomes. Outer track: local LD support ($ld_w > 0.2$) per marker; inner ring: the genome-wide sorting classification.

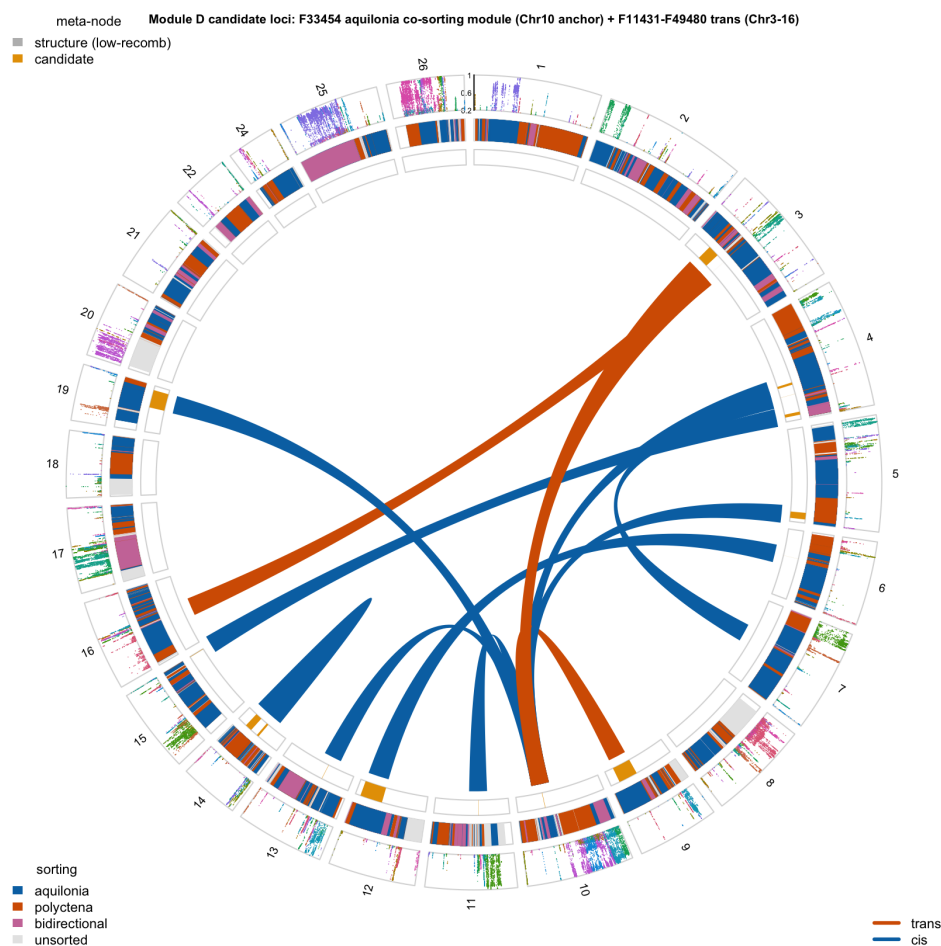


Figure 5: **Candidate associations on the genome.** Each association drawn as a band spanning the two meta-nodes it joins (cis blue, trans orange; full opacity, minimum width for visibility). The Chr10 anchor F33454 fans out to aquilonia-sorted partners on Chr3/4/5/9/11/13/19 (the aquilonia co-sorting module), while the single orange band F11431–F49480 (Chr3–Chr16) is the one ancestry-independent trans pair.